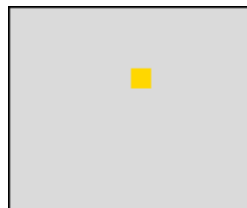


Instructions Treasure Hunt Task

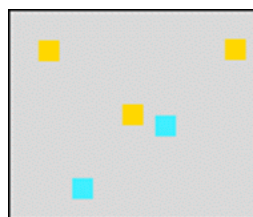
In this task, you are a miner digging for treasures. These are **blue zircons** and **gold nuggets**. Your mining company sends to mines in different countries where you'll have to dig for these treasures. All of the mines contain both treasures, but some of them will have more gold nuggets, others more blue zircons. You can only either dig out either the zircons or the gold. Because both of them are of the same value, it is important that you dig out the treasure which is more plentiful, in order to maximize your returns.



When you arrive at a mine, you obviously don't know whether there is more gold or blue zircons. But you have a little helper that assists you: a robot will randomly dig in a part of the mine and report back the treasure that it found. Here, for example, gold was found in this location:



What happens during the game is that the robot will go down, dig in 5 new, random places and report back to you what it found. Here 2 blue zircons and 3 gold in two locations:

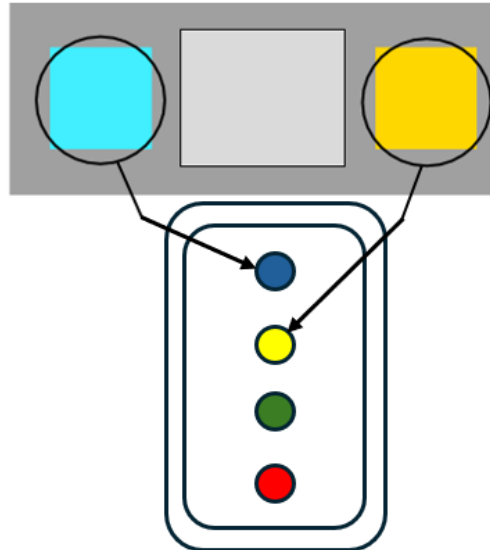


The location of the dug treasure is purely random and does not contain any useful information. Moreover, the robot will never dig the same area twice, so each new piece of information is novel. All the treasures will have the same size, it will always dig out 5 new pieces, but they can be in different colours.

The robot go down into the mine, report back to you, will go down again, report back to you, and so on. So you will see new information about every second.

The robot will keep on uncovering coloured areas **until you are certain enough** whether there will be more gold nuggets or blue zircons in this mine. Once you are certain enough, you will decide whether you want dig for gold or the zircon by pressing the button which represents this treasure.

When you have made your decision, you have to press the golden or blue button. The colour which is assigned with each button will be displayed on the screen. The colour patches can switch the position in different mines, so please pay attention to which colour is where.

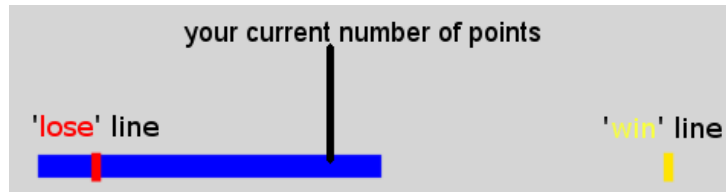


You can wait as long as you want with making your decision. However, at some point, the robot runs out of batteries and obstructs the mine. If that happens, you cannot decide anymore and dig out any treasure. So make sure that you decide before it runs out of battery. When you pressed the button, you cannot change your decision anymore.

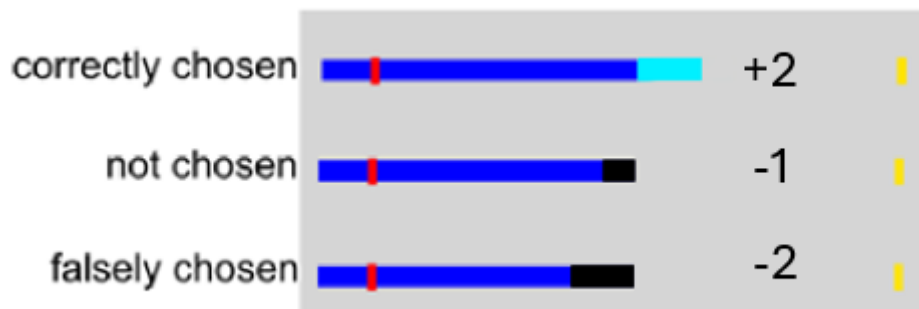
Importantly, the battery life is always about the same, so you will get a feeling in the practice session on how long it will be working, although there is some variability. When the robot runs out of battery, you will see that the map will turn pink and you can no longer decide for a colour:



You as a mine digger will get (in addition to your hourly payment) paid based on your performance. At the bottom of the screen, you will see the following bar indicating your performance. In blue, you will see how many points you currently have acquired. Your aim is to cross the yellow line. Every time you hit the yellow line, you will get additional €1, paid to you at the end of the experiment. Of course, once you hit the line, you will start over in the middle. However, if you are not successful, you may hit the red line. Then, you will have to pay €1 to the mining company.



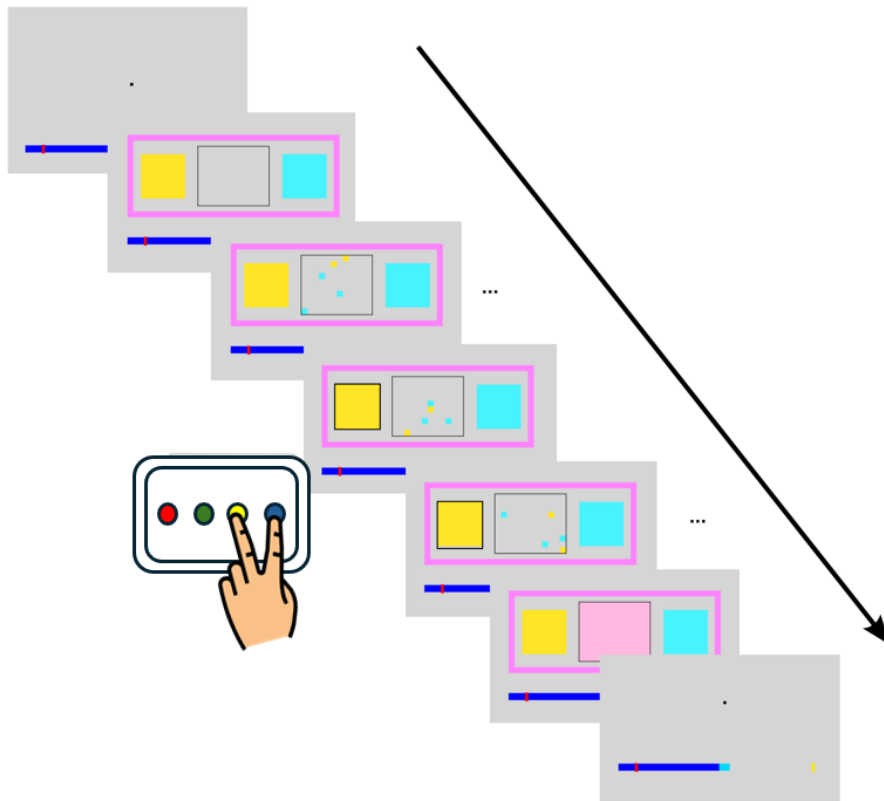
After every mine, your number of points (i.e. the blue bar) will be updated. If you choose the colour which is truly more plentiful, you will get 2 points and your bar will grow (top). If you choose the wrong colour, then you lose 2 points and the blue bar will turn black and shrink (bottom). If you did not decide until the robot ran out of battery, you will lose 1 point and the blue bar will also shrink a bit, but not as much as if you chose the wrong colour (middle).



For practice only: Please note that the robot keeps on digging once you have made your decision until it runs out of battery. This way you will get a feeling for how long each robot battery will last.

Every mine is independent of each other and they do not influence each other.

Just a quick recap [explain the game again]:

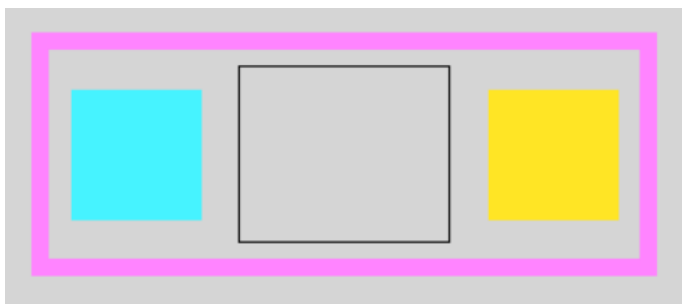


[practice 1: task_v08_MEGEEG > open TreasHunt_pract_main.m > press F5 > enter subject number and practice number (1)]

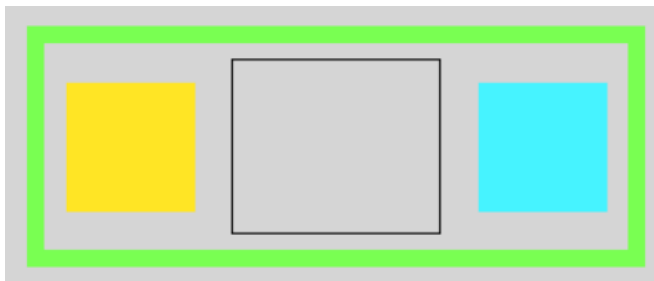
Is all clear? In the real game, there is one more thing you need to pay attention to:

In the mines in which you will be working, there will be two different types of robots. They will not differ in the way that they dig the treasures. The only difference is in their battery life: One robot type has a longer battery life than the other. Thus, with the robot which has the longer battery life, you will be able to gather more information before making the decision. Also here, the battery life is stable across the different mines, with only slight variation. Moreover, it is **exactly the same as in the next practice**. So you will already have a feeling for how long you can wait when you enter the scanner. In order that you know which type of robot you are operating, you will see a colour frame around the mine and buttons. One of the robots has a green colour, the other has a pink colour. Please take this into account when you make your decision. **[tell subject which colour is short/long: uneven subjects: pink=short; even subjects: green=short]**

Pink robot:



Green robot:



[practice 2: as above]

We are almost ready to begin the experiment. You will play 4 blocks of ~15 minutes each, with breaks between the blocks. Please keep in mind that the colour frames indicate how much time you've got.

Please remain still during the whole experiment and try not to move too much, because this may lead to data loss.

Between the blocks, we'll talk to you, ask how you are and you can relax a bit.

MRI (done by Tobias):

- Movement is bad! Do not move your head while doing the task – during the breaks it's fine